

#s: 1, 4, 6, 8

Math 3210 — Chapter 1, section 1

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1: If $a, b \in \mathbb{R}$ and $a < b$, give a description in set theory notation for each of the intervals (a, b) , $[a, b]$, $[a, b)$ and $(a, b]$.

$$(a, b) = \{x \in \mathbb{R} : a < x < b\}; \quad (a, b] = \{x \in \mathbb{R} : a < x \leq b\}$$
$$[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}; \quad [a, b) = \{x \in \mathbb{R} : a \leq x < b\}$$

4: What is the intersection of all open intervals containing the closed interval $[0, 1]$? Justify your answer.

Say $a, b \in \mathbb{R}$. Find $\bigcap_{\substack{a < 0 \\ 1 < b}} (a, b)$

Conjecture: $\bigcap_{\substack{a < 0 \\ 1 < b}} (a, b) = [0, 1]$, because the intersection of all sets that have $[0, 1]$ is at least $[0, 1]$ but isn't more in this case.

Proof: $\bigcap_{\substack{a < 0 \\ 1 < b}} (a, b)$ is the same as $\{x : a < x < b \ \forall \ a < 0, \ 1 < b\}$

Say we find a max 'a' value. Then $(a_{\max} + 0)/2$ produces a new $a_{\max}$. $\forall a \in \mathbb{R}$ this holds true. Thus if $x > a$, then the 'x least' value is 0.

Likewise for a least 'b' value, $(1 + b_{\min})/2$ produces a new $b_{\min}$. Thus 'x most' is 1.
So, $0 \leq x \leq 1$ which is $x \in [0, 1]$.

Say $x \in [0, 1]$. Then $\{x : 0 \leq x \leq 1\}$.

Then x is certainly greater than a if $a < 0$.
So $a < x \ \forall \ a < 0$.

Likewise x is certainly less than b if $b > 1$.
So $b > x \ \forall \ b > 1$.

Thus $\{x : a < x < b \ \forall \ a < 0, \ 1 < b\}$.

Then x is certainly contained within the intersection

$\bigcap_{\substack{a < 0 \\ 1 < b}} (a, b)$.

□

Ch 1.1, #s 6, 8

6: What is the union of all closed intervals contained in the open interval $(0,1)$? Justify your answer.

Conjecture: $\bigcup \{S_n\} \subset (0,1) = (0,1)$

Proof: Since the union can be written as $\{x: x \in S_1 \text{ or } x \in S_2 \text{ or } \dots\}$ which must all be contained within $(0,1)$.

So $S_1, S_2, S_3, \dots$ are all subsets of $(0,1)$ and are all closed intervals.

It therefore suffices to show $\nexists x \in S_n$ s.t. $x \leq 0$ or $x \geq 1$.

But any set S which contained such an x would not be a subset of $(0,1)$ which is to say it would not be contained within the interval $(0,1)$.

So $\forall x \in S_n, x \in (0,1)$.

Say $y \in (0,1)$; then $\{y: 0 < y < 1\}$.

If $\exists y: 0 < y < 1$ then certainly $y \in [x, z]$ where x was always some value greater than 0 evidenced by $(0+x)/2$.

Likewise y is always some value less than 1 evidenced by $(y+1)/2$.

So y would definitely be in the union of all $[x, y]$. So $y \in \bigcup_{\substack{x > 0 \\ 1 > y}} [x, y]$. $\square$

8. Which of the following functions $f: \mathbb{R} \rightarrow \mathbb{R}$ are one-to-one and which ones are onto? Justify your answer.
 a) $f(x) = x^2$ b) $f(x) = x^3$ c) $f(x) = e^x$

One-to-One:

b
c

Onto:

b

a) $f(x) = x^2$; $\forall f(x), 0 \leq f(x)$ and is thus not onto as $\nexists f(x) = -1$. Since $f: \mathbb{R} \rightarrow \mathbb{R}$, the co-domain is $\mathbb{R}$ and the image of f is $[0, \infty)$.
 Also, $\exists x, y \in \mathbb{R}$ s.t. $x \neq y$ and $f(x) = f(y)$, so it is not one-to-one.

b) $f(x) = x^3$; Since one-to-one is defined as "distinct inputs map to distinct outputs", there are no Real inputs that get mapped to more than once from Real inputs. $f(x_1) = f(x_2) \Rightarrow x_1^3 = x_2^3 \Rightarrow x_1^3 - x_2^3 = 0 \Rightarrow (x_1 - x_2)(x_1^2 + x_1x_2 + x_2^2) = 0 \Rightarrow x_1 = x_2$.
 Since we are mapping from $\mathbb{R} \rightarrow \mathbb{R}$, is all of $\mathbb{R}$ represented? By the fundamental theorem of algebra there will always be one real and two complex roots. Therefore all of $\mathbb{R}$ is represented under the image of f .

c) $f(x) = e^x$. $f(x) = e^x$ is one-to-one because e^x is differentiable everywhere (by definition) which it wouldn't be were there repeated values of the co-domain, by definition. $f(x) = e^x$ is not, however, onto because f maps from $\mathbb{R}$ to $\mathbb{R} - \{0\}$, meaning the image is less than the set $\mathbb{R}$.

#s: 2, 3, 10, 13

Tyler Trotter

Homework 1, Section 1.2

2. Prove that if $n, m \in \mathbb{N}$, then $m+n \neq n$. Use induction on n .

Proposition: $m+n \neq n$ where $n, m \in \mathbb{N}$

Suppose we fix $m \in \mathbb{N}$.

Base Case P_1 : $m+1 \neq 1$ is always true because $m+1$ is the definition of $S(m)$ and by N_3 1 is not the successor of any $m \in \mathbb{N}$.

Inductive Step: Suppose P_n is true for some n . Then $m+n \neq n$. Then P_{n+1} is $m+(n+1) \neq n+1$.

LHS: By 1.2.4 $m+(n+1) = m+S(n) = S(m+n)$

RHS: By 1.2.3 $n+1 = S(n)$

So, $m+S(n) \neq S(n)$ is a true statement because there is no element $m \in \mathbb{N}$ that satisfies the additive identity. Thus, by induction, P_n is true for all $n \in \mathbb{N}$. $\square$

3. Use (2) to prove that if $n, m \in \mathbb{N}$, $n \leq m$ and $m \leq n$ then $n=m$.

Say $n \leq m$ and $m \leq n$ is true. Then we say:
1) $(n < m \text{ or } n=m)$ and $(m < n \text{ or } n=m)$ which is to say
 $(m = n+x \text{ or } n=m)$ and $(n = m+y \text{ or } n=m)$, for some $x, y \in \mathbb{N}$.
If $m = n+x$ and $n = m+y$ were true, then $m = m+y+x$, which is false for all $x, y \in \mathbb{N}$ by (2).

2) Likewise for $n = n+x+y$.

3) Say $n=m$ and $n = m+y$ were true. Then $m = m+y$ which is impossible by (2).

4) Likewise for $m = n+x$ and $n=m$.

So, say $n=m$ and $m=n$. This is true and holds for any $n, m \in \mathbb{N}$.

Thus $n=m$. $\square$

#s. 10, 13

Homework 1, Section 1.2 (cont.)

10: Using induction, prove that $\sum_{k=1}^n (2k-1) = n^2$ for every $n \in \mathbb{N}$.

Base Case: $(2(1)-1) = 1^2$ is true.

Inductive Step: Say P_n is true for some $n \in \mathbb{N}$.

$$\begin{array}{ccccccc} [(1) + (3) + \dots + (2n-1)] + (2(n+1)-1) \\ \begin{array}{ccccccc} 1 & 2 & \dots & n & & n+1 \\ n^2 + (2(n+1)-1) = n^2 + 2n + 1 = (n+1)^2 \end{array} \end{array}$$

Thus $\sum_{k=1}^n (2k-1) = n^2$ is true for all $n \in \mathbb{N}$. $\square$

13: Let a sequence $\{x_n\}$ of numbers be defined recursively by $x_1 = 0$ and $x_{n+1} = \frac{x_n + 1}{2}$.

Prove by induction that $x_n \leq x_{n+1}$ for all $n \in \mathbb{N}$.

Say P_n is $x_n \leq x_{n+1}$.

Base Case, P_1 : $x_2 = \frac{0+1}{2} = \frac{1}{2}$, so $x_1 \leq x_2$ is true.

Inductive Step, P_{n+1} : Say P_n is true for some $n \in \mathbb{N}$.

Then $x_n \leq x_{n+1}$, where $x_{n+1} = \frac{x_n + 1}{2}$.

$$\text{So, } \frac{x_n + 1}{2} \leq \frac{x_{n+1} + 1}{2}$$

Which is to say $x_{n+1} \leq x_{n+2}$.

So, $x_n \leq x_{n+1}$ is true for all $n \in \mathbb{N}$. $\square$

Would this hold true for $x_1 = 2$?

Base Case, P_1 : $x_2 = \frac{2+1}{2} = \frac{3}{2}$. $x_1 = 2 \leq \frac{3}{2} = x_2$ is false.

Thus, since the Base Case is false, the equality is false when $x_1 = 2$.